DIEP TUONG NGHIEM

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EDUCATION

VNUHCM - University of Science, Ho Chi Minh City, VN — 09/2021 - 11/2025
Bachelor of Science - Advanced Program in Computer Science — **GPA**: 9.39/10.0, 3.92/4.0

VNUHCM-High School for the Gifted, Ho Chi Minh City, VN — 08/2018 - 06/2021
Student in Specialized Math Class — **GPA**: 8.9/10.0

EXPERIENCE

VinRobotics | Ha Noi, VN — 01/2026 - 07/2026
AI Research Engineer | Supervisor: An LE, Duy NGUYEN

- Research topics: Spatial Intelligence in Vision-Language-Action Models, World Models for Robotics, Vision-Language-Action Models Pre-Training.
- Responsibility: Lead research projects, Discuss and propose the algorithm, Implement algorithm, Code for downstream tasks.

VinUniversity | Ha Noi, VN — 05/09/2025 - 30/11/2025
Research Assistant | Supervisor: Khoa DOAN

- Research topics: Vision-Language-Action Models, Spatial Intelligence, Data Efficiency.
- Responsibility: Discuss and propose the algorithm, Implement algorithm, Code for downstream tasks, Prepare dataset (download and preprocess).

Application Driven Mathematics-VinBigData | Ha Noi, VN — 07/2024 - 05/2025
Research Assistant | Supervisor: Nhat HO

- Research topics: PEFT, Foundation Model, LLM, Multi-Modal
- Responsibility: Discuss and propose the algorithm, Implement algorithm.

MBZUAI | UAE — 27/05/2024 - 21/06/2024
Research Internship | Supervisor: Mohsen Guizani

- Research topics: Federated Learning for Brain Tumor Detection.
- Responsibility: Propose the algorithm, Build variations of Federated Learning for brain tumor detection.

German Research Center for Artificial Intelligence | Germany — 03/2023 - 07/2026
Research Assistant | Supervisor: Duy NGUYEN

- Research topics: Self-Supervised Learning for 2D/3D Medical Image and Multi-Modal tasks, Optimal Transport applications, Multi-Modal Foundation Model for Medical, Graph Learning.
- Responsibility: Discuss and propose the algorithm, Implement algorithm, Code for downstream tasks, Prepare dataset (download and preprocess).

AISIA Lab | Ho Chi Minh City, VN — 02/2022 - 09/2025
Member | Supervisor: Binh NGUYEN, Duy NGUYEN

- Research topics: Image-Text Retrieval for Lifelogging, Audio Captioning.
- Responsibility: Lead teams, Discuss and propose the algorithm, Implement algorithm.

RESEARCH PUBLICATION

Selected Publications

1. Minh Duc Nguyen*, Nghiem Tuong Diep*, Nguyen Gia Binh*, Trong-Bao Ho, Doanh Le Thien, Quang Tan Nguyen, Thien-Loc Ha, Tran Van Nhiem, Bao Thach, Tran Xuan Nhat, Tuan Anh Tran, Artur Habuda, Philip Lund Møller, Tran Nguyen Le, Daniel Sonntag, Mathias Niepert, Khoa D Doan, Vu N. Duong, Hung Ngo, Minh Nhat VU, Duy Minh Ho Nguyen, An Thai Le**, Vien Anh Ngo**. **FOCA: Future-Oriented Conditioning for Data-Efficient Vision-Language-Action Adaptation.** *International Conference on Machine Learning (ICML) 2026.* Link

2. Nghiem T. Diep*, Huy Nguyen*, Chau Nguyen*, Minh Le, Duy M. H. Nguyen, Daniel Sonntag, Mathias Niepert, Nhat Ho. **On Zero-Initialized Attention: Optimal Prompt and Gating Factor Estimation.** *International Conference on Machine Learning (ICML) 2025.* Link

3. Duy M. H. Nguyen, Nghiem T. Diep*, Trung Q. Nguyen*, Hoang-Bao Le, Tai Nguyen, Tien Nguyen, TrungTin Nguyen, Nhat Ho, Pengtao Xie, Roger Wattenhofer, James Zhou, Daniel Sonntag, Mathias Niepert. **EXGRA-MED: Extended Context Graph Alignment for Medical Vision- Language Models.** *Advances in Neural Information Processing Systems (NeurIPS) 2025.* Link

4. Anh-Tien Nguyen, Duy Minh Ho Nguyen*, Nghiem T. Diep*, Trung Quoc Nguyen, Nhat Ho, Jacqueline Michelle Metsch, Miriam Cindy Maurer, Daniel Sonntag, Hanibal Bohnenberger, Anne-Christin Hauschild. **MG-PATH: Vision-Language Model with Multi-Granular Prompt Learning for Few-Shot WSI Classification.** *Transactions on Machine Learning Research (TMLR) 2025.* Link

5. Thinh Pham*, Nghiem Diep*, Lizi Liao, Binh Nguyen. **DistillCaps: Enhancing Audio-Language Alignment in Captioning via Retrieval-Augmented Knowledge Distillation.** *The Conference on Information and Knowledge Management (CIKM) 2025.* Link

6. Duy MH Nguyen*, An T Le*, Trung Q Nguyen, Nghiem T Diep, Tai Nguyen, Duy Duong-Tran, Jan Peters, Li Shen, Mathias Niepert, Daniel Sonntag. **Dude: Dual Distribution-Aware Context Prompt Learning For Large Vision-Language Model.** *Asian Conference on Machine Learning (ACML) 2024.* Link

7. Duy MH Nguyen, Hoang Nguyen, Nghiem T Diep, Tan N Pham, Tri Cao, Binh T Nguyen, Paul Swoboda, Nhat Ho, Shadi Albarqouni, Pengtao Xie, Daniel Sonntag, Mathias Niepert. **LVM-Med: Learning Large-Scale Self-Supervised Vision Models for Medical Imaging via Second-order Graph Matching.** *Advances in Neural Information Processing Systems (NeurIPS) 2023.* Link

AWARDS

- **MediaEval 2022** AI Challenge: **"Distinctive Mention" Awardee** in Urban Air Quality task.
- **SHREC 2023** AI Challenge: **2nd place** in TextANIMAR: Text-based 3D animal fine-grained retrieval track.
- **Finalist** for AWS-Build On Vietnam (2021).
- **First prize** for SOMC - Southern Open Mathematics Competition (2019).
- **Second prize** for the Mathematics Competition for Excellent Student of Major High Schools in the Northern Delta and Coastal Areas (2019).
- **Silver award** in VANDA International Science Competition.
- **Consolation prize** in Viet Nam Mathematical Olympiad (2020).

SKILLS

- ***Computer Science Background***: Computer Science (C++, Python), Mathematics, Machine Learning, Deep Learning, Computer Vison.
- ***Research Interests:*** Vision-Language-Action Models, Spatial Intelligence, World Models, Graph Learning, Medical Multi-Modal Foundation Models, Parameter-Efficient Fine-Tuning.
- ***Languages***: English (7.0 IELTS), Vietnamese (native speaker).
- ***Programming Languages***: Python, C++, Deep Learning libraries (PyTorch, Numpy).

REFERENCES

1. Assoc. Prof. Pengtao Xie (Email: p1xie@ucsd.edu)
   Google Scholar: https://scholar.google.com/citations?user=cnncomYAAAAJ&hl=en
   Associate Professor at University of California San Diego, USA.

2. Prof. Dr.-Ing. Daniel Sonntag (Email: sonntag@dfki.de)
   Google Scholar: https://scholar.google.com/citations?user=v7i6Uz4AAAAJ&hl=en
   German Research Center for Artificial Intelligence, DFKI.

3. Assoc. Prof. Nguyen Thanh Binh (Email: ngtbinh@hcmus.edu.vn)
   Google Scholar: https://scholar.google.com.co/citations?user=dXEb3PMAAAAJ&hl=en
   Associate Professor at University of Science, Vietnam.

4. Assoc. Prof. Ho Pham Minh Nhat (Email: minhnhat@utexas.edu)
   Google Scholar: https://scholar.google.com/citations?user=Xs7cKMwAAAAJ&hl=en
   Associate Professor at University of Texas at Austin, USA.

5. PhD Candidate Nguyen Ho Minh Duy (Email: nhmduyhcmus@gmail.com)
   Google Scholar: https://scholar.google.com/citations?user=_NIyeykAAAAJ&hl=en
   PhD Candidate at Max Planck Research School for Intelligent Systems & DFKI, Germany.